THE MORROW–PATTERSON LEBESGUE FUNCTION IS NOT CENTRALLY SYMMETRIC AT $n = 2$

Abstract. Beberok, Białas-Cież and De Marchi study the Lebesgue function λ_n of polynomial interpolation at the Morrow–Patterson points $\mathrm{MP}_n \subset Q = [-1,1]^2$ for n a positive even integer. They prove $\lambda_n(x,y) = \lambda_n(-x,y)$, and in their final remarks they pose the proof of $\lambda_n(x,y) = \lambda_n(-x,-y)$ as an open problem. That identity is false at $n = 2$: the point $Q_0 = \left(0, \cos\frac{2\pi}{5}\right)$ has

$$\lambda_2(Q_0) = 2 - \frac{1}{\sqrt{5}} = 1.552786404500042061\ldots, \qquad \lambda_2(-Q_0) = 1,$$

the second value holding because $-Q_0$ is an interpolation node. Both are exact elements of $\mathbb{Q}(\sqrt{2},\sqrt{5})$ and the refutation reduces to $2-1/\sqrt{5} \neq 1$; in fact $\lambda_2 > 1$ at all six points of $-\mathrm{MP}_2$. We also show, for every even n, that $-\mathrm{MP}_n$ is disjoint from MP_n and that the two sets partition the full tensor grid, so central symmetry in degree n would force one $\binom{n+2}{2} \times \binom{n+2}{2}$ integer-indexed kernel matrix to be entrywise nonnegative; we prove that implication in this direction only, and make no claim about a converse. At $n = 2$ that matrix has a negative entry equal to -2. The failure of central symmetry is proved here only at $n = 2$.

1. The identity in question

Let n be a positive even integer, $Q = [-1,1]^2$, and let U_j denote the Chebyshev polynomial of the second kind, $U_0 = 1$, $U_1(t) = 2t$, $U_{j+1}(t) = 2tU_j(t) - U_{j-1}(t)$. The *Morrow–Patterson points* [2] of degree n are

(1)
$$\mathrm{MP}_n = \left\{(x_m, y_k)\ :\ 1 \le m \le n+1,\ 1 \le k \le \tfrac{n}{2}+1\right\}, \qquad x_m = \cos\frac{m\pi}{n+2},$$

with

$$y_k = \cos\frac{2k\pi}{n+3}\ \ (m \text{ odd}), \qquad y_k = \cos\frac{(2k-1)\pi}{n+3}\ \ (m \text{ even}),$$

in the form used in [1, §3.2]. Then $\#\,\mathrm{MP}_n = \frac{(n+1)(n+2)}{2} = \dim \mathcal{P}^2_n$ and MP_n is unisolvent for total-degree-n bivariate polynomials, so the Lebesgue function

$$\lambda_n(x,y) = \sum_{P \in \mathrm{MP}_n} \left|L_P(x,y)\right|$$

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is defined, L_P being the Lagrange basis. Beberok, Białas-Cież and De Marchi [1] write it as

$$\lambda_n(x,y) = \sum_{(x_m,y_k)\in \mathrm{MP}_n} \omega_{m,k} \Big| \sum_{i+j\le n} U_i(x_m)U_j(y_k)\, U_i(x)U_j(y)\Big|, \tag{2}$$

where the cubature weights are given both by

$$\frac{1}{\omega_{m,k}} = \sum_{i+j\le n} U_i^2(x_m)U_j^2(y_k) \tag{3}$$

and, in closed form, by $\omega_{m,k} = C_n(1-x_m^2)(1-y_k^2)$ with $C_n = \frac{8}{(n+2)(n+3)}$ [1, Thm. 2]. Write

$$K_n(P,Q) = \sum_{i+j\le n} U_i(x_P)U_j(y_P)\, U_i(x_Q)U_j(y_Q), \qquad L_P = \omega_P\, K_n(P,\cdot). \tag{4}$$

The reflection identity

$$\lambda_n(x,y) = \lambda_n(-x,y) \qquad ((x,y)\in Q) \tag{5}$$

is [1, Lemma 1], and [1] closes with the following passage, its *Final remarks*, §7.1 (quoted from the e-print arXiv:2412.14595v1):

> "The numerical evidence shows that the Lebesgue function, with the notation of the paper $\lambda(x,y)$, is symmetric not only for y, but also to the origin. By Lemma 18 above we proved that $\lambda(x,y) = \lambda(-x,y)$. However, it is an open problem of proving that $\lambda(x,y) = \lambda(-x,-y)$. Another open problem is to prove that the Lebesgue constant is attained at each corner of the square as suggested by the numerical results."

The identity proved there is (5); the second of the two open problems is not considered here. The identity $\lambda(x,y) = \lambda(-x,-y)$ is posed in the notation of [1], in which n is a positive even integer, and it is set against (5), which is proved there for every such n. The degree $n = 2$ is among the degrees at issue in [1]: the set MP_2 is displayed in its Figure 2, and the Lebesgue constants evaluated there run over $n = 2, 4, \ldots, 60$.

Theorem 1. *The identity $\lambda_n(x,y) = \lambda_n(-x,-y)$ fails at $n = 2$. Explicitly, let $n = 2$ and*

$$Q_0 = \left(0,\ \cos\tfrac{2\pi}{5}\right) = \left(0,\ \tfrac{\sqrt{5}-1}{4}\right).$$

Then $-Q_0 \in \mathrm{MP}_2$, whence $\lambda_2(-Q_0) = 1$, while

$$\lambda_2(Q_0) = 2 - \frac{1}{\sqrt{5}} = \frac{10-\sqrt{5}}{5} = 1.552786404500042061\ldots \neq 1.$$

Since $2-1/\sqrt{5} = 1$ would force $\sqrt{5} = 1$, one exact evaluation in $\mathbb{Q}(\sqrt{2},\sqrt{5})$ refutes the identity; no decimal approximation enters the argument.

2. The counterexample

Throughout this section $n = 2$, so $\dim \mathcal{P}_2^2 = 6$, and we abbreviate

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad a = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2},$$

$$c_1 = \cos\frac{\pi}{5} = \frac{\varphi}{2} = \frac{\sqrt{5}+1}{4}, \qquad c_2 = \cos\frac{2\pi}{5} = \frac{\varphi-1}{2} = \frac{\sqrt{5}-1}{4}.$$

Only $\varphi^2 = \varphi + 1$ and $a^2 = \frac{1}{2}$ are used below. From (1), $x_1 = a$, $x_2 = 0$, $x_3 = -a$; the odd columns $m = 1, 3$ carry $y \in \{\cos\frac{2\pi}{5}, \cos\frac{4\pi}{5}\} = \{c_2, -c_1\}$ and the even column $m = 2$ carries $y \in \{\cos\frac{\pi}{5}, \cos\frac{3\pi}{5}\} = \{c_1, -c_2\}$, so

$$\mathrm{MP}_2 = \{P_1, \ldots, P_6\},$$

$$\begin{aligned} &P_1 = \left(\tfrac{\sqrt{2}}{2}, \tfrac{\sqrt{5}-1}{4}\right), \quad P_2 = \left(\tfrac{\sqrt{2}}{2}, -\tfrac{\sqrt{5}+1}{4}\right), \quad P_3 = \left(0, \tfrac{\sqrt{5}+1}{4}\right), \\ &P_4 = \left(0, \tfrac{1-\sqrt{5}}{4}\right), \qquad P_5 = \left(-\tfrac{\sqrt{2}}{2}, \tfrac{\sqrt{5}-1}{4}\right), \quad P_6 = \left(-\tfrac{\sqrt{2}}{2}, -\tfrac{\sqrt{5}+1}{4}\right). \end{aligned} \tag{6}$$

Lemma 2. $-Q_0 = \left(0, \frac{1-\sqrt{5}}{4}\right)$ *is the node* (x_2, y_2) *of* MP_2*, and* $Q_0 \notin \mathrm{MP}_2$*. Consequently* $\lambda_2(-Q_0) = 1$*, with no computation.*

Proof. $m = 2$ is even, so the $m = 2$ column carries $y_k = \cos\frac{(2k-1)\pi}{5}$; at $k = 2$ this is $\cos\frac{3\pi}{5} = -\cos\frac{2\pi}{5}$, which is the second coordinate of $-Q_0$, and the first coordinate is $x_2 = \cos\frac{\pi}{2} = 0$. That column carries only $y \in \{c_1, -c_2\}$ and $\cos\frac{2\pi}{5} = c_2$ is neither, while every other column has $x = \pm a \neq 0$; hence $Q_0 \notin \mathrm{MP}_2$. A Lebesgue function equals 1 at each interpolation node. $\square$

By (3), or equally from the closed form with $C_2 = \frac{2}{5}$, the six weights of (6) are

$$(\omega_{P_1}, \ldots, \omega_{P_6}) = \left(\tfrac{2+\varphi}{20}, \tfrac{3-\varphi}{20}, \tfrac{3-\varphi}{10}, \tfrac{2+\varphi}{10}, \tfrac{2+\varphi}{20}, \tfrac{3-\varphi}{20}\right), \tag{7}$$

and $\sum_P \omega_P = 1$ exactly.

Proof of Theorem 1. At $Q_0 = (0, c_2)$ the kernel (4) collapses: $U_1(0) = 0$ and $U_2(0) = -1$, so every term with $i = 1$ vanishes and, using $U_1(c_2) = \varphi - 1$ and $U_2(c_2) = (\varphi-1)^2 - 1 = 1 - \varphi$,

$$K_2(P, Q_0) = 1 + (\varphi - 1)\left[U_1(y_P) - U_2(y_P)\right] - U_2(x_P) \qquad (P \in \mathrm{MP}_2). \tag{8}$$

Evaluating (8) on (6), with $U_2(\pm a) = 1$, $U_2(0) = -1$ and $\varphi^2 = \varphi + 1$:

$$\begin{array}{llll} P_1: & U_1 - U_2 = 2(\varphi - 1), & U_2(x) = 1, & K_2 = 2(2-\varphi) = 3 - \sqrt{5}, \\ P_2: & U_1 - U_2 = -2\varphi, & U_2(x) = 1, & K_2 = -2\varphi(\varphi-1) = -2, \\ P_3: & U_1 - U_2 = 0, & U_2(x) = -1, & K_2 = +2, \\ P_4: & U_1 - U_2 = 0, & U_2(x) = -1, & K_2 = +2, \\ P_5: & \text{as } P_1, & & K_2 = 3 - \sqrt{5}, \\ P_6: & \text{as } P_2, & & K_2 = -2, \end{array}$$

abbreviating $U_1 - U_2$ for $U_1(y_P) - U_2(y_P)$ and $U_2(x)$ for $U_2(x_P)$. Exactly two of the six values are negative, both equal to -2, at P_2 and P_6, whose common weight is $\omega = \frac{3-\varphi}{20} = \frac{5-\sqrt{5}}{40}$.

The Lagrange functions reproduce constants, so that $\sum_P L_P \equiv 1$, i.e. $\sum_P \omega_P K_2(P, Q) = 1$ for every Q; here

$$\sum_P \omega_P K_2(P, Q_0) = 2 \cdot \tfrac{2+\varphi}{20}(4 - 2\varphi) - 2 \cdot \tfrac{3-\varphi}{20} \cdot 2 + \tfrac{3-\varphi}{10} \cdot 2 + \tfrac{2+\varphi}{10} \cdot 2 = 1.$$

Splitting $|t| = t + 2t^-$ with $t^- = \max(-t, 0)$ therefore gives

$$\begin{aligned} \lambda_2(Q_0) &= \sum_P \omega_P \big|K_2(P, Q_0)\big| = 1 + 2 \sum_{K_2(P,Q_0)<0} \omega_P\big(-K_2(P, Q_0)\big) \\ &= 1 + 2 \cdot \Big(2 \cdot \tfrac{5-\sqrt{5}}{40} \cdot 2\Big) = 1 + 8 \cdot \tfrac{5-\sqrt{5}}{40} = 1 + \frac{5-\sqrt{5}}{5}, \end{aligned}$$

that is

$$\lambda_2(Q_0) = 2 - \frac{\sqrt{5}}{5} = 2 - \frac{1}{\sqrt{5}} = \frac{10 - \sqrt{5}}{5} = 1.552786404500042061\ldots$$

With Lemma 2, $\lambda_2(Q_0) - \lambda_2(-Q_0) = 1 - \frac{\sqrt{5}}{5} \neq 0$. $\square$

At $n = 2$ the identity fails at every one of the six antipodal node pairs. Indeed $\lambda_2 \equiv 1$ on MP_2, while negating (6) pointwise and evaluating (2) in $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ gives

$$\begin{gathered} \lambda_2(-P_1) = \lambda_2(-P_5) = \tfrac{5}{2} - \tfrac{3\sqrt{5}}{10}, \qquad \lambda_2(-P_2) = \lambda_2(-P_6) = 2 + \tfrac{3\sqrt{5}}{5}, \\ \lambda_2(-P_3) = 2 + \tfrac{\sqrt{5}}{5}, \qquad \lambda_2(-P_4) = \lambda_2(Q_0) = 2 - \tfrac{\sqrt{5}}{5}, \end{gathered} \tag{9}$$

none of which is 1; the value at Q_0 is the smallest of the six.

3. The parity mechanism, for every even n

Theorem 1 is an evaluation. Behind it is a parity count, which holds in every even degree.

Index a point of the full tensor grid by (m, jj), meaning $\big(\cos\frac{m\pi}{p}, \cos\frac{jj\pi}{q}\big)$ with

$$p = n + 2 \text{ (even)}, \qquad q = n + 3 \text{ (odd)}, \qquad 1 \le m \le p - 1, \quad 1 \le jj \le q - 1.$$

In this indexing (1) reads

$$\mathrm{MP}_n = \big\{(m, jj) \ : \ jj \text{ even} \iff m \text{ odd}\big\}, \tag{10}$$

and the antipodal map $(x, y) \mapsto (-x, -y)$ is $\sigma : (m, jj) \mapsto (p - m,\, q - jj)$.

Proposition 3. *For every even $n \ge 2$:*

1. $\sigma_x : (m, jj) \mapsto (p - m, jj)$ *maps* MP_n *onto itself, because p is even and so $m \mapsto p - m$ preserves the parity of m. This is* (5).

(2) *$jj \mapsto q - jj$ reverses the parity of jj, because q is odd. Hence $\sigma(\mathrm{MP}_n) = -\mathrm{MP}_n$ is the set defined by the opposite parity rule to* (10)*: it is* disjoint *from* MP_n*, and* $\mathrm{MP}_n \sqcup (-\mathrm{MP}_n)$ *is the whole grid, of cardinality* $(n+1)(n+2)$.

(3) *$\lambda_n(-x, -y) = \lambda_n(x, -y)$ for all $(x, y) \in Q$. Hence the identity $\lambda_n(x, y) = \lambda_n(-x, -y)$ on Q holds if and only if λ_n is symmetric about the x-axis.*

(4) *Since $L_P = \omega_P K_n(P, \cdot)$ and $\sum_P L_P \equiv 1$, one has $\lambda_n \geq 1$ everywhere, with equality at Q if and only if $K_n(P, Q) \geq 0$ for every $P \in \mathrm{MP}_n$. As $\lambda_n \equiv 1$ on MP_n, the identity $\lambda_n(x, y) = \lambda_n(-x, -y)$ forces $\lambda_n \equiv 1$ on $-\mathrm{MP}_n$, hence forces the $D \times D$ matrix $\big[K_n(P, Q)\big]_{P \in \mathrm{MP}_n,\, Q \in -\mathrm{MP}_n}$, $D = \frac{(n+1)(n+2)}{2}$, to be entrywise nonnegative. A single negative entry refutes it.*

Proof. (1) and (2) are the parity statements just made, and the cardinality $\#\,\mathrm{MP}_n = \frac{(n+1)(n+2)}{2}$ is half the grid, which is what (2) asserts. (3) is (5) with y replaced by $-y$. For (4), $\sum_P L_P \equiv 1$ gives $\lambda_n(Q) = \sum_P |L_P(Q)| \geq \big|\sum_P L_P(Q)\big| = 1$, with equality exactly when no $L_P(Q)$ is negative, and $\omega_P > 0$. □

We use Proposition 3(4) in one direction only: one negative entry of that matrix refutes central symmetry in degree n. We neither prove nor claim a converse, so the criterion is a necessary condition and not a finite reduction of the identity. At $n = 2$ Theorem 1 exhibits a negative entry: in the column $Q = Q_0$ of that 6×6 matrix two of the six entries equal -2, and 14 of its 36 entries are negative.

Proposition 3 holds in every even degree, but the failure of central symmetry is proved here only at $n = 2$; we do not claim that $\lambda_n(x, y) = \lambda_n(-x, -y)$ fails for any $n \geq 4$. A reader who takes the passage quoted in Section 1 to concern only the large degrees at which the Lebesgue function is plotted in [1] should treat that reading as untouched by Theorem 1.

4. Verification

Every quantity displayed above lies in $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ and was recomputed independently in exact arithmetic. The accompanying program `verify.py` reads the printed objects — the node set (6), the weights (7), the kernel column of the proof of Theorem 1 and the values (9) — as literal strings, and rederives each of them from (1), (2), (3) and (4). It represents an element of $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ by its four rational coordinates in the basis $1, \sqrt{2}, \sqrt{5}, \sqrt{10}$, so equality is decided coordinatewise; the only order decisions, those inside the absolute values of (2), are made by bracketing the element between rational bounds obtained from integer square roots and refining until the bracket excludes 0. No floating-point number decides anything. The program also confirms the cosine values used in Section 2 by the Chebyshev identity $T_5(\cos \frac{k\pi}{5}) = (-1)^k$ together with the ordering, checks unisolvence of MP_2 through $\omega_i K_2(P_i, P_j) = \delta_{ij}$ on all 36 pairs, and verifies Proposition 3(1),(2) as integer statements about (10) for every even n from 2 to 40. It uses the Python standard library only, prints one line per check, and reports

```
VERDICT: ALL 39 CHECKS PASS
```

Its transcript accompanies this note. Five of those checks concern quantities not claimed here: two further forms of the weights (7), and the values $\lambda_2(1,1)$ and $\lambda_2(-1,-1)$ together with their difference, which belong to the separate corner-attainment problem of the passage quoted in Section 1. The program is not needed in order to redo the computation: the proof of Theorem 1 is five lines of arithmetic in φ, and every input it consumes is printed above.

References

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[2] C. R. Morrow and T. N. L. Patterson, *Construction of algebraic cubature rules using polynomial ideal theory*, SIAM J. Numer. Anal. **15** (1978), 953–976.